VISIONINSPECT AI

MILESTONE 4 — TESTING, DEPLOYMENT & DOCUMENTATION

PREPARED BY:-

RUCHIRA NAYAK

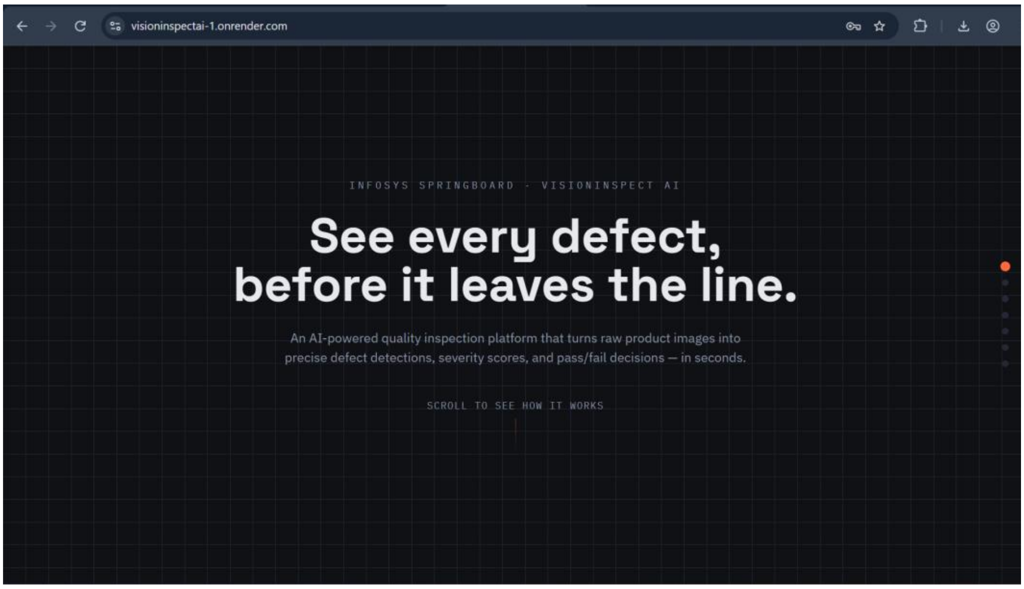
Infosys Springboard Internship

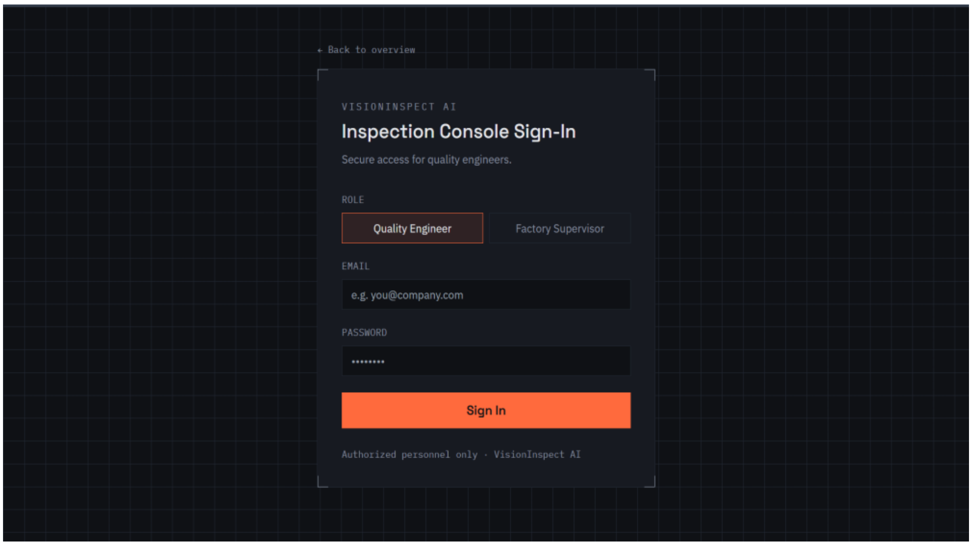
Milestone – 4

1. Milestone Overview

Milestone 4 focused on the final testing, integration, deployment, and documentation of **VisionInspect AI**, an AI-powered manufacturing quality inspection platform. The objective of this milestone was to bring together the modules developed during the previous milestones and validate the complete inspection workflow.

During this phase, the image preprocessing, anomaly detection, defect classification, severity assessment, analytics, authentication, database, inspection history, and reporting components were integrated and tested. The platform was also prepared for cloud deployment and final demonstration.



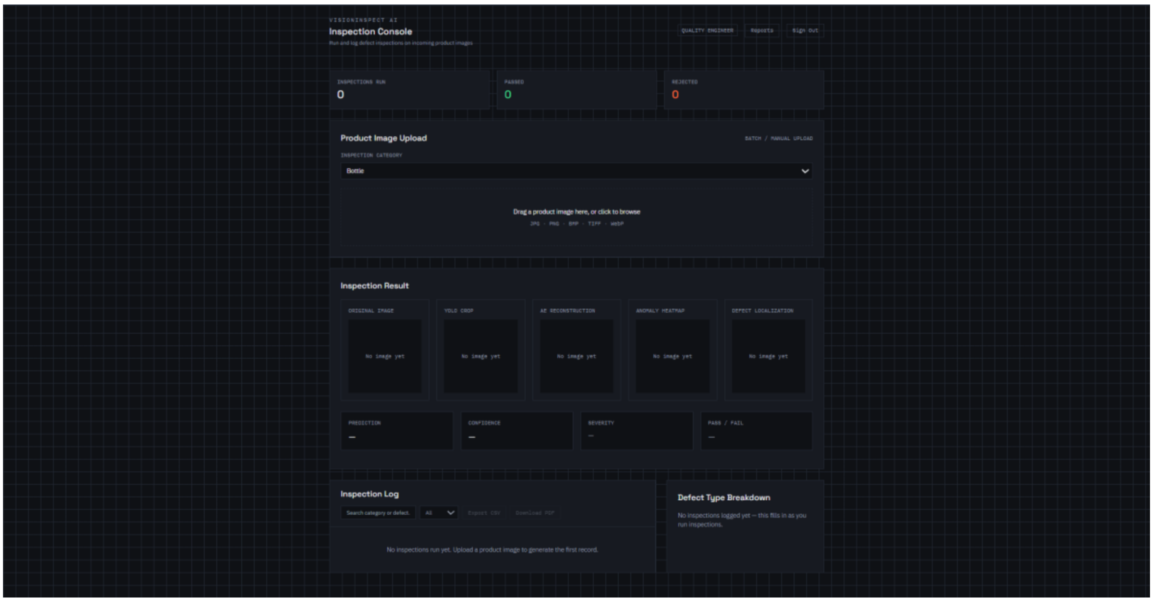


2. Final System Integration

The individual components developed throughout the project were integrated into a unified manufacturing inspection system. The final application allows an authenticated user to upload a product image, select or identify the relevant product category, perform AI-based inspection, and receive a detailed quality assessment.

The backend integrates authentication, image upload, dataset handling, preprocessing, augmentation, severity analysis, statistics, and inspection management functionality. This integration enables the different modules to work together as a single end-to-end application.

The final system therefore moves beyond individual model experimentation and provides a complete workflow suitable for demonstrating an industrial visual inspection process.



3. AI-Based Inspection Pipeline

The final VisionInspect AI inspection pipeline consists of multiple stages designed to progressively analyse a manufacturing product image.

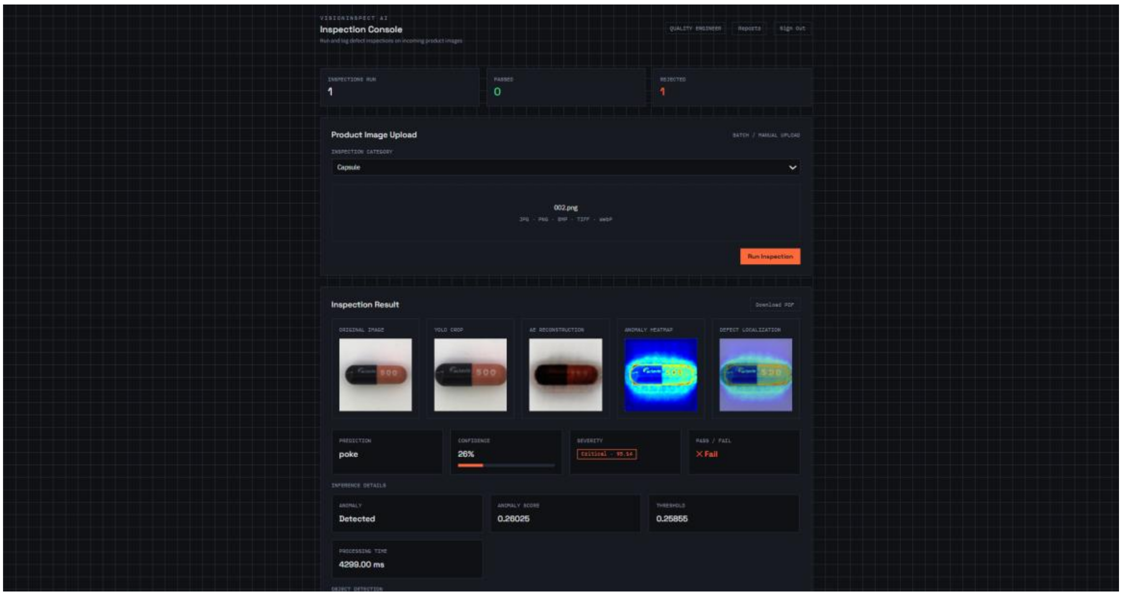
The process begins with image preprocessing and quality validation. Where applicable, YOLO-based object detection is then used to identify the relevant product region. The cropped image is analysed using an autoencoder-based anomaly detection model, followed by SSIM-based anomaly-map generation.

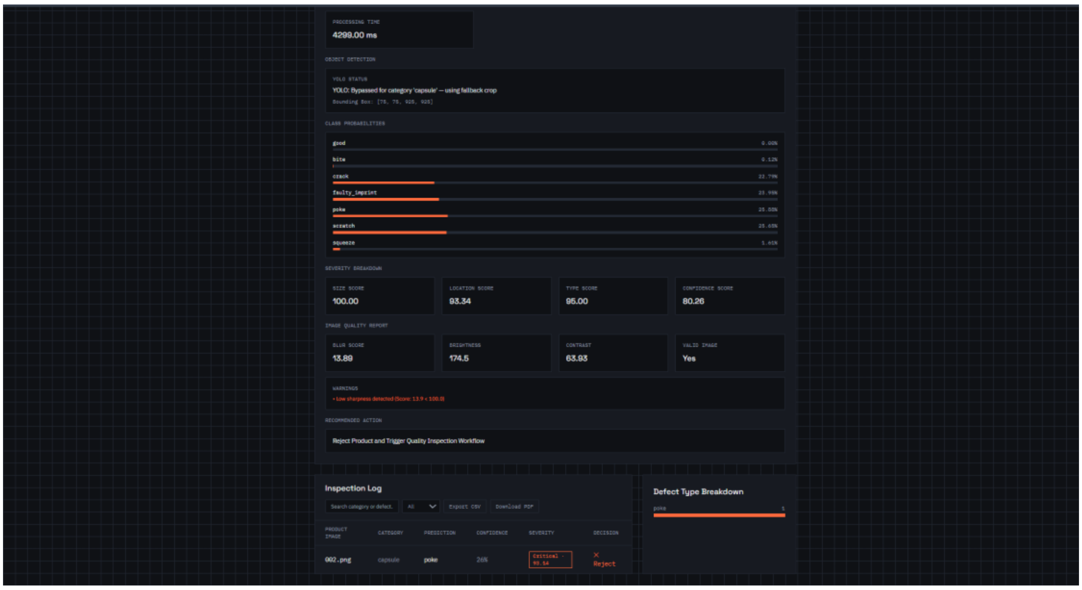
The resulting anomaly information is used to determine whether the product passes inspection or should be rejected. For anomalous products, the system performs defect classification and severity assessment. Finally, visual outputs such as heatmaps and overlays are generated to make the detected defect easier to interpret.

The complete pipeline therefore follows:

Image Input → Preprocessing → Product Detection → Anomaly Detection → PASS/REJECT Decision → Defect Classification → Severity Assessment → Visualization

This integrated sequence represents the core AI functionality of VisionInspect AI.



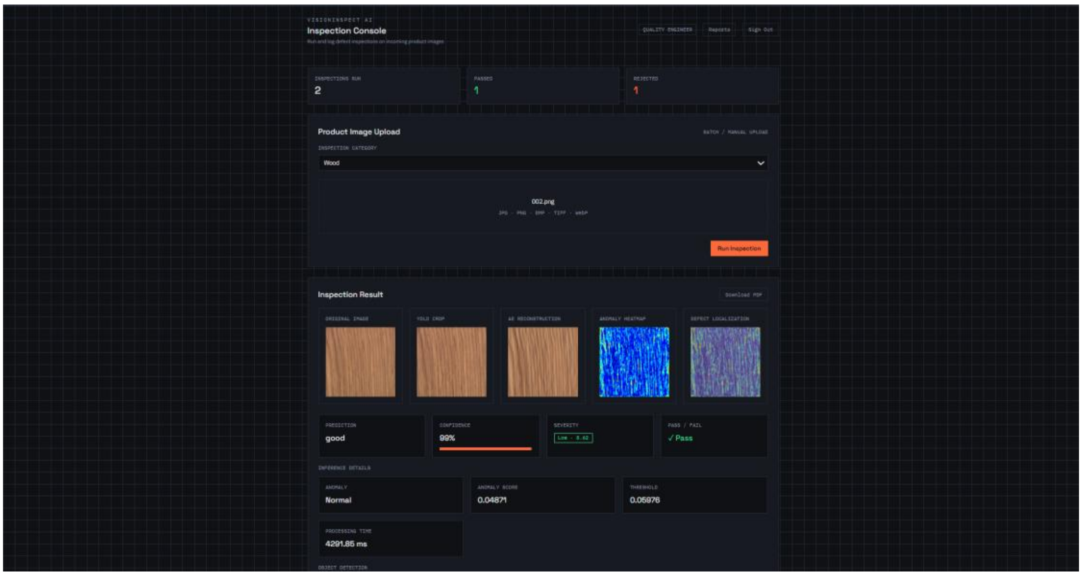


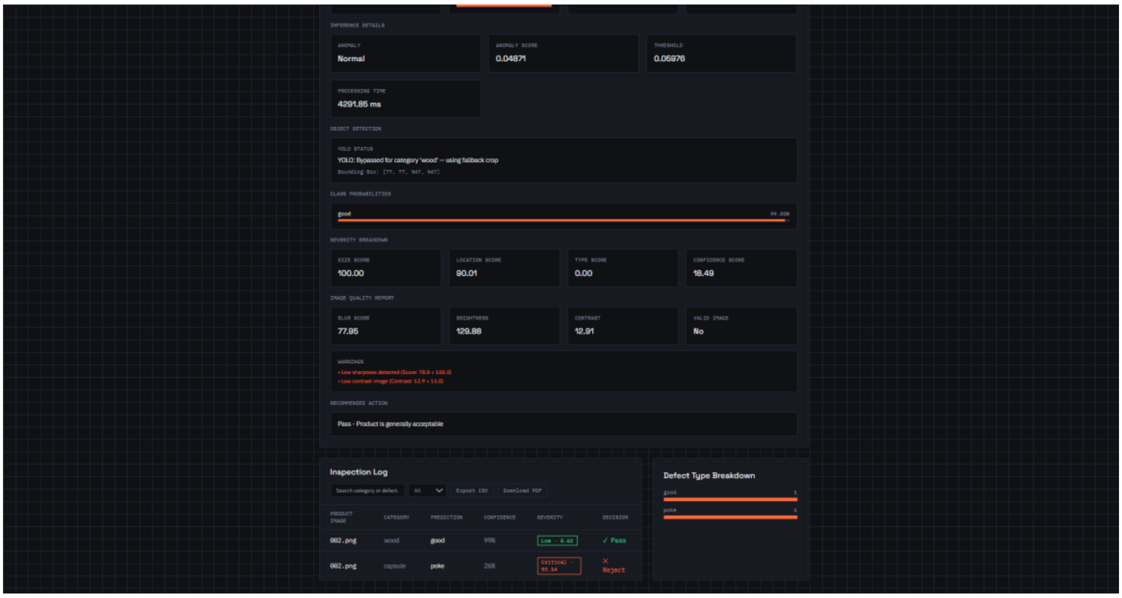
4. Model Testing and Validation

The AI inspection pipeline was tested using product images to verify the behaviour of the different stages of the system. Testing focused on image preprocessing, model loading, anomaly detection, defect classification, decision generation, and visual output generation.

The system supports category-specific model loading, allowing the appropriate trained model to be selected according to the product category being inspected. This approach provides a scalable structure for handling multiple manufacturing product categories.

The prediction workflow was also tested through the backend API to ensure that an uploaded image could successfully pass through the complete inference pipeline and return an inspection result.





5. Defect Visualization

An important part of the final system is the visualization of detected anomalies.

Instead of providing only a textual prediction, VisionInspect AI generates visual information indicating the regions where abnormal patterns are detected. Heatmaps and overlays help users understand the location and approximate extent of the detected defect.

This improves the interpretability of the AI system and allows an inspector to visually verify the areas highlighted by the model.

6. Severity Assessment and Quality Decision

The severity assessment module was integrated into the final inspection workflow to provide more meaningful quality-control information.

The system considers factors including defect size, defect location, defect type, and model confidence while calculating the severity score. The resulting score is mapped to different severity levels, allowing defects to be categorized according to their potential impact on product quality.

The final assessment can therefore provide not only a defect prediction but also an indication of whether the defect represents a low, medium, high, or critical quality concern.

This information can assist quality-control personnel in deciding whether a product should be accepted, reviewed, repaired, reworked, or rejected.

7. Inspection History and Analytics

VisionInspect AI maintains inspection-related information to support historical analysis and manufacturing quality monitoring.

The system records inspection information and provides statistical insights such as the number of inspected products, good products, defective products, pass rate, and failure rate.

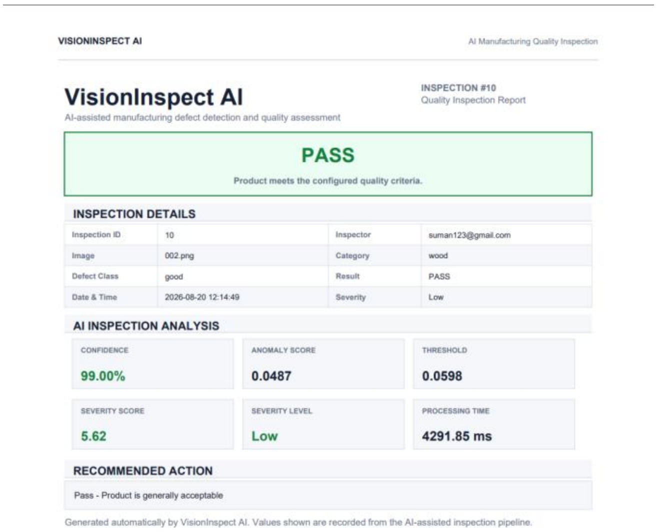
The analytics functionality allows users to monitor inspection outcomes and identify quality trends over time. This transforms the system from a simple defect-detection tool into a broader manufacturing quality-monitoring platform.

8. Automated Inspection Reports

The reporting functionality was integrated into the final inspection workflow to provide structured documentation of inspection results.

An inspection report can contain important information including the inspected product, inspection result, defect classification, confidence score, anomaly score, severity score, severity level, processing information, and recommended action.

This provides a permanent record of an inspection and can be useful for quality-control documentation, production monitoring, and future analysis.


VISIONINSPECT AI

AI Manufacturing Quality Inspection

VisionInspect AI

AI-assisted manufacturing defect detection and quality assessment

INSPECTION #10

Quality Inspection Report

PASS

Product meets the configured quality criteria.

INSPECTION DETAILS

Inspection ID	10	Inspector	suman123@gmail.com
Image	002.png	Category	wood
Defect Class	good	Result	PASS
Date & Time	2026-08-20 12:14:49	Severity	Low

AI INSPECTION ANALYSIS

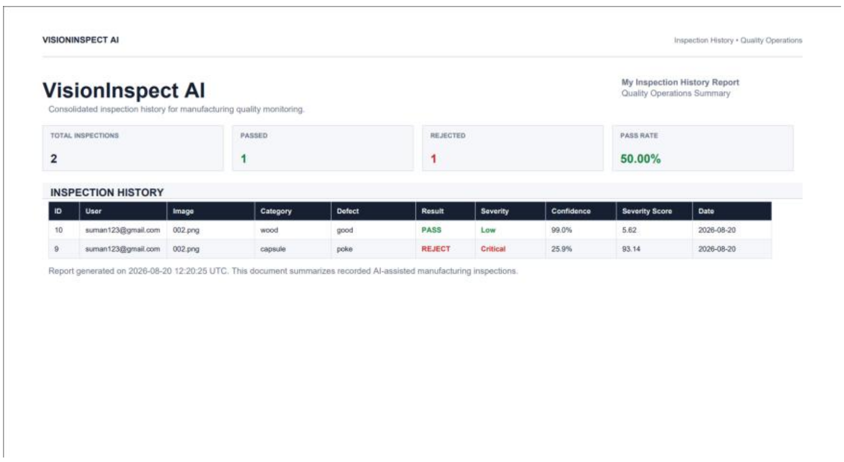
CONFIDENCE	ANOMALY SCORE	THRESHOLD
99.00%	0.0487	0.0598

SEVERITY SCORE	SEVERITY LEVEL	PROCESSING TIME
5.62	Low	4291.85 ms

RECOMMENDED ACTION

Pass - Product is generally acceptable

Generated automatically by VisionInspect AI. Values shown are recorded from the AI-assisted inspection pipeline.


VISIONINSPECT AI

Inspection History • Quality Operations

VisionInspect AI

Consolidated inspection history for manufacturing quality monitoring.

My Inspection History Report

Quality Operations Summary

TOTAL INSPECTIONS	PASSED	REJECTED	PASS RATE
2	1	1	50.00%

INSPECTION HISTORY

ID	User	Image	Category	Defect	Result	Severity	Confidence	Severity Score	Date
10	suman123@gmail.com	002.png	wood	good	PASS	Low	99.0%	5.62	2026-08-20
9	suman123@gmail.com	002.png	capsule	poke	REJECT	Critical	25.9%	93.14	2026-08-20

Report generated on 2026-08-20 12:20:25 UTC. This document summarizes recorded AI-assisted manufacturing inspections.

9. Factory Supervisor Dashboard

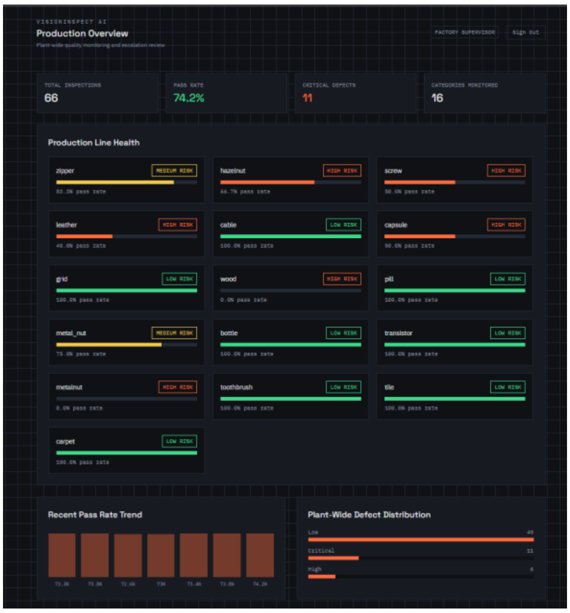
The Factory Supervisor Dashboard provides a centralized view of manufacturing inspection activities and quality performance. It is designed specifically for supervisors to monitor inspection results and identify quality issues without having to inspect individual records manually.

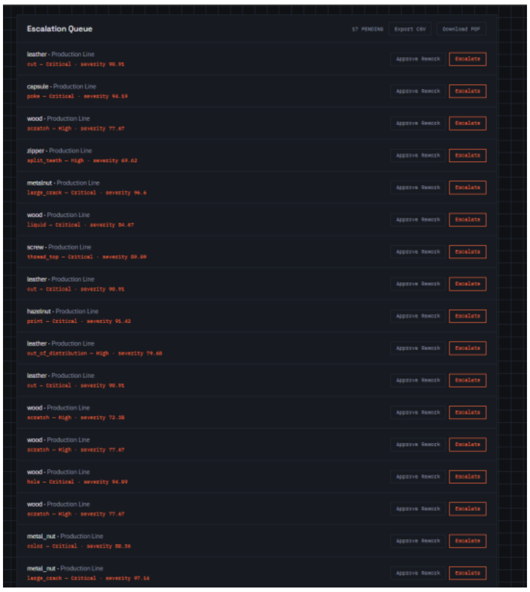
The dashboard provides an overview of important manufacturing quality indicators, including:

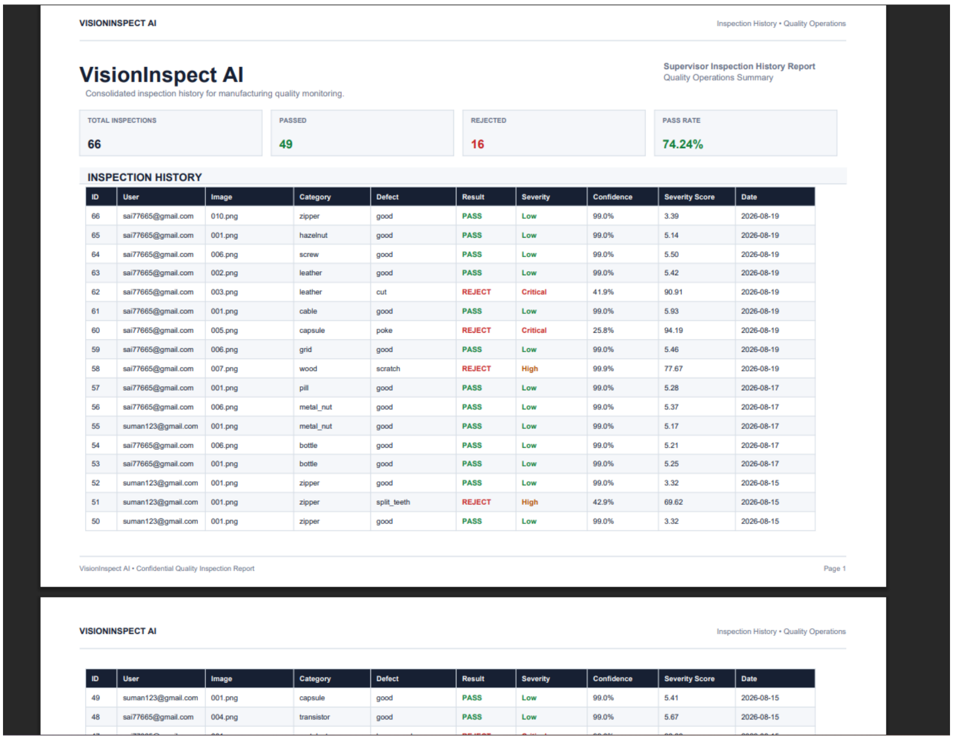
- Total inspections performed
- Passed and rejected products
- Defect distribution
- Defect severity levels
- Inspection trends
- Product/category-wise quality performance
- Recent inspection activity

The dashboard helps factory supervisors quickly identify recurring defects, monitor rejection patterns, and understand the overall quality status of the production process. It provides a higher-level view of the inspection data generated by the AI system and supports faster quality-control decision-making.

The supervisor can use these insights to identify products or categories requiring additional attention, investigate recurring manufacturing defects, and monitor the effectiveness of the inspection process.







VISIONINSPECT AI

Inspection History • Quality Operations

VisionInspect AI

Consolidated inspection history for manufacturing quality monitoring.

Supervisor Inspection History Report
Quality Operations Summary

TOTAL INSPECTIONS	PASSED	REJECTED	PASS RATE
66	49	16	74.24%

INSPECTION HISTORY

ID	User	Image	Category	Defect	Result	Severity	Confidence	Severity Score	Date
66	sai77665@gmail.com	010.png	zipper	good	PASS	Low	99.0%	3.39	2026-08-19
65	sai77665@gmail.com	001.png	hazelnut	good	PASS	Low	99.0%	5.14	2026-08-19
64	sai77665@gmail.com	006.png	screw	good	PASS	Low	99.0%	5.50	2026-08-19
63	sai77665@gmail.com	002.png	leather	good	PASS	Low	99.0%	5.42	2026-08-19
62	sai77665@gmail.com	003.png	leather	cut	REJECT	Critical	41.9%	90.91	2026-08-19
61	sai77665@gmail.com	001.png	cable	good	PASS	Low	99.0%	5.93	2026-08-19
60	sai77665@gmail.com	005.png	capsule	poke	REJECT	Critical	25.8%	94.19	2026-08-19
59	sai77665@gmail.com	006.png	grid	good	PASS	Low	99.0%	5.46	2026-08-19
58	sai77665@gmail.com	007.png	wood	scratch	REJECT	High	99.9%	77.67	2026-08-19
57	sai77665@gmail.com	001.png	pill	good	PASS	Low	99.0%	5.28	2026-08-17
56	sai77665@gmail.com	006.png	metal_nut	good	PASS	Low	99.0%	5.37	2026-08-17
55	suman123@gmail.com	001.png	metal_nut	good	PASS	Low	99.0%	5.17	2026-08-17
54	sai77665@gmail.com	006.png	bottle	good	PASS	Low	99.0%	5.21	2026-08-17
53	sai77665@gmail.com	001.png	bottle	good	PASS	Low	99.0%	5.25	2026-08-17
52	suman123@gmail.com	001.png	zipper	good	PASS	Low	99.0%	3.32	2026-08-15
51	suman123@gmail.com	001.png	zipper	split_teeth	REJECT	High	42.9%	69.62	2026-08-15
50	suman123@gmail.com	001.png	zipper	good	PASS	Low	99.0%	3.32	2026-08-15

VisionInspect AI • Confidential Quality Inspection Report

Page 1

VISIONINSPECT AI

Inspection History • Quality Operations

ID	User	Image	Category	Defect	Result	Severity	Confidence	Severity Score	Date
49	suman123@gmail.com	001.png	capsule	good	PASS	Low	99.0%	5.41	2026-08-15
48	sai77665@gmail.com	004.png	transistor	good	PASS	Low	99.0%	5.67	2026-08-15

10. Cloud Deployment

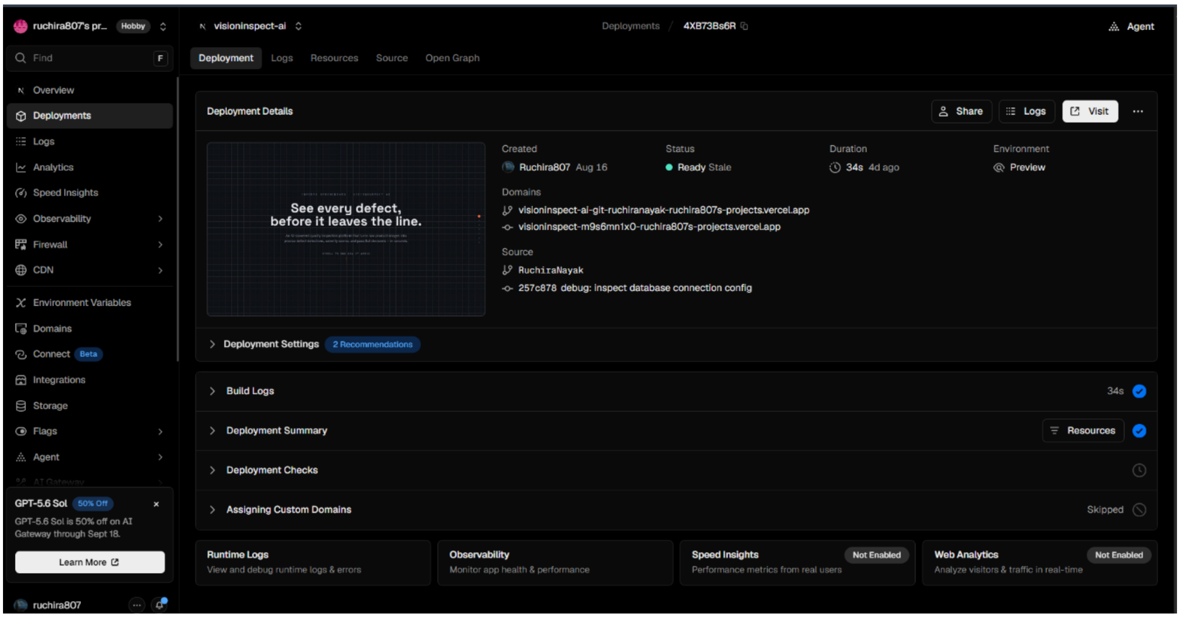
The final application was prepared for cloud deployment by separating the frontend, backend, and database into their respective services.

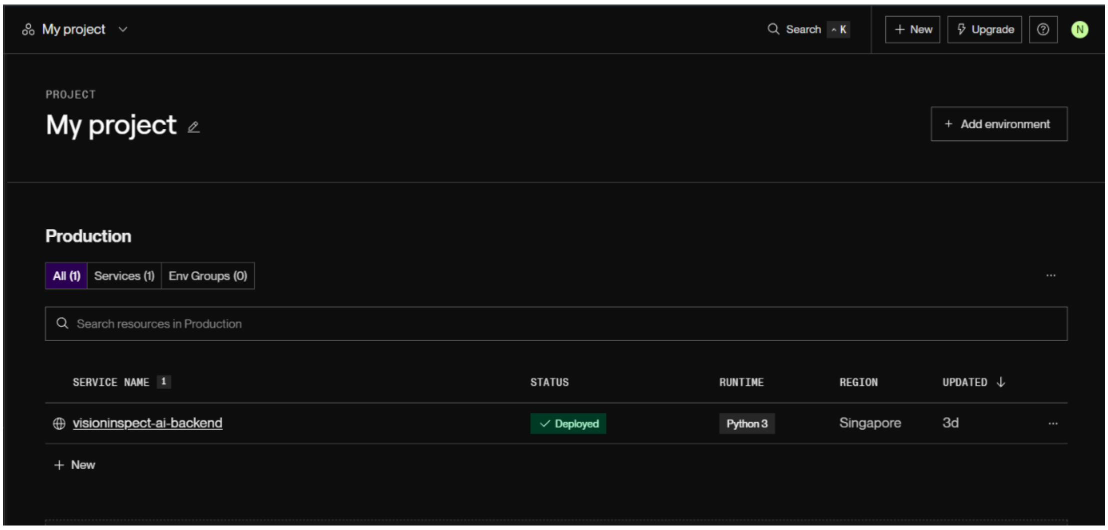
The deployment architecture consists of:

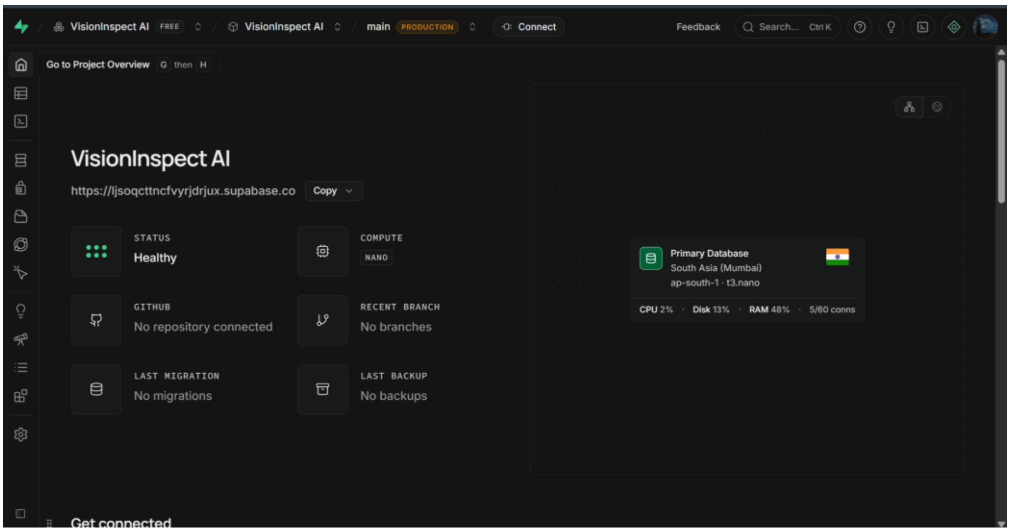
Frontend — Vercel
Backend — Render
Database — Supabase
Source Code — GitHub

This architecture allows the frontend interface to communicate with the remotely hosted FastAPI backend while the backend communicates with the production database.

The backend was configured to allow requests from the deployed frontend environment, enabling communication between the cloud-hosted components.


ruchira807's pr... Hobby
visioninspect-ai
Deployments / 4XB73Bs6R
Agent
Find
Deployment
Logs
Resources
Source
Open Graph
Overview
Deployments
Logs
Analytics
Speed Insights
Observability
Firewall
CDN
Environment Variables
Domains
Connect Beta
Integrations
Storage
Flags
Agent
GPT-5.6 Sol 50% Off
GPT-5.6 Sol is 50% off on AI Gateway through Sept 18.
Learn More
ruchira807
Deployment Details
Share
Logs
Visit
See every defect, before it leaves the line.
Created
Ruchira807 Aug 16
Status
Ready Stale
Duration
34s 4d ago
Environment
Preview
Domains
visioninspect-ai-git-ruchiranayak-ruchira807s-projects.vercel.app
visioninspect-m9s6mn1x0-ruchira807s-projects.vercel.app
Source
RuchiraNayak
257c878 debug: inspect database connection config
Deployment Settings
2 Recommendations
Build Logs
34s
Deployment Summary
Resources
Deployment Checks
Assigning Custom Domains
Skipped
Runtime Logs
View and debug runtime logs & errors
Observability
Monitor app health & performance
Speed Insights
Not Enabled
Performance metrics from real users
Web Analytics
Not Enabled
Analyze visitors & traffic in real-time


My project
Search
New
Upgrade
PROJECT
My project
Add environment
Production
All (1)
Services (1)
Env Groups (0)
Search resources in Production
SERVICE NAME
STATUS
RUNTIME
REGION
UPDATED
visioninspect-ai-backend
Deployed
Python 3
Singapore
3d
New


VisionInspect AI FREE
VisionInspect AI
main PRODUCTION
Connect
Feedback
Search... Ctrl K
Go to Project Overview G then H
VisionInspect AI
https://ljsoqcttncfvyrjdrjux.supabase.co
Copy
STATUS
Healthy
COMPUTE
NANO
GITHUB
No repository connected
RECENT BRANCH
No branches
LAST MIGRATION
No migrations
LAST BACKUP
No backups
Primary Database
South Asia (Mumbai)
ap-south-1 · t3.nano
CPU 2%
Disk 13%
RAM 48%
5/60 conns
Get connected

11. End-to-End System Testing

The complete VisionInspect AI workflow was tested from the user interface through the backend and AI model pipeline.

The final workflow consists of:

User Login → Image Upload → Product Category Selection → Image Processing → AI Inspection → Anomaly Detection → Defect Classification → Severity Assessment → Result Visualization → Database Storage → Analytics/Report

This testing verified the integration between the frontend, backend, AI inference pipeline, and database.

The successful completion of the workflow demonstrates that the individual modules developed during the earlier milestones can operate together as a unified manufacturing inspection platform.

12. Conclusion and Learning Outcomes

Milestone 4 completed the final testing, integration, deployment, and documentation phase of VisionInspect AI.

The project now provides an integrated AI-based manufacturing inspection workflow capable of processing product images, detecting anomalies, classifying defects, assessing severity, generating visual inspection results, maintaining inspection records, and providing quality analytics.

The milestone provided practical experience in AI model validation, computer vision pipelines, backend API development, frontend-backend integration, database management, cloud deployment, API testing, system integration, and technical documentation.

Overall, VisionInspect AI demonstrates the application of artificial intelligence and computer vision techniques to automate and improve manufacturing quality inspection processes.